AI-Based Multi-Department Helpdesk System for Public Sector Governance Using Retrieval-Augmented Generation and Role-Based Escalation

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Abstract— Public sector organizations and government departments administer extensive procedural workflows spanning payroll, taxation, pension processing, information technology access, and general administration. Regulatory rules governing these workflows are conventionally disseminated across heterogeneous office circulars, gazettes, and service manuals. As a consequence, administrative personnel and employees face recurrent information bottlenecks, leading to excessive telephone inquiries, email backlogs, and duplicated administrative overhead. This paper proposes a secure, multi-departmental intelligent helpdesk architecture integrating document-grounded Retrieval-Augmented Generation (RAG) with an automated human-in-the-loop ticketing workflow. Approved department circulars are segmented via sliding-window chunking, mapped into dense semantic vectors using sentence transformers, and indexed within an isolated vector database. Query answering is conducted by retrieving top-K contextual chunks and prompting an on-premise, open-source Large Language Model (LLM) to synthesize factual, source-attributed responses. To eliminate hallucination risks, an anti-hallucination threshold dynamically escalates low-confidence or unindexed inquiries to departmental officers via an integrated ticketing pipeline. The system implements four-tier Role-Based Access Control (RBAC) enforced through JSON Web Tokens (JWT) and bcrypt cryptographic hashing. Rigorous empirical validation demonstrates an average retrieval confidence score of 0.78, an 85% positive user feedback rating, 100% test-set hallucination prevention, and sub-second query latency under lexical fallback. The complete architecture is containerized and publicly deployed across global edge infrastructure.

Keywords— *Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Helpdesk Automation, Role-Based Access Control, Government Information Systems, Vector Embeddings, ChromaDB, FastAPI.*

I. INTRODUCTION

Public administration and governmental secretariats operate under strict regulatory codices, service conduct rules, and administrative manuals. Within such state and central bureaucratic environments, government employees routinely encounter procedural ambiguities concerning Dearness Allowance (DA) revisions, leave travel concessions (LTC), Income Tax declarations (Form 12BB), National Pension System (NPS) superannuation, and National Informatics Centre (NIC) credential management. Traditionally, resolving these standard queries necessitates physical desk visits, manual circular cross-referencing, or raising ad-hoc email requests, creating substantial delays and consuming valuable administrative human resources.

While conversational artificial intelligence has advanced rapidly with the advent of Large Language Models (LLMs), off-the-shelf commercial conversational agents present critical vulnerabilities when deployed in government institutions. First, foundation models trained on open-web corpora are inherently prone to stochastic confabulation or 'hallucinations'—generating legally invalid or fabricated policies with high linguistic confidence [11]. Second, third-party cloud-hosted AI APIs violate institutional sovereign data-residency and employee confidentiality protocols. Third, isolated automated chatbots frequently become dead-ends when encountering out-of-scope or unprecedented inquiries, lacking structural bridges to human grievance mechanisms.

To resolve these technical and operational deficiencies, this paper presents an end-to-end, multi-department enterprise helpdesk portal designed specifically for public administration. The core contributions of this study are: (1) An isolated, multi-departmental Retrieval-Augmented Generation (RAG) engine that restricts dense semantic retrieval to verified institutional documents, outputting grounded natural-language answers with strict document-level source citations; (2) An integrated anti-hallucination guardrail that evaluates cosine similarity bounds (theta = 0.35 and tau = 0.60), triggering an automated transition to human-in-the-loop ticket resolution when confidence is inadequate; (3) A four-tier Role-Based Access Control (RBAC) system providing customized operational views for Employees, Officers, Department Admins, and Super Administrators; and (4) Comprehensive empirical benchmarking across local environments and cloud edge

infrastructure (Vercel, Render, and Supabase), demonstrating operational reliability across 33/33 validated test suites.

II. LITERATURE REVIEW

A. Evolution of Conversational Agents and Information Retrieval

Early rule-based dialogue systems, exemplified by ELIZA [1], relied upon hardcoded pattern matching and decomposition rules. Although deterministic, rule-based systems suffer from catastrophic brittleness when handling unscripted user phrasing, synonym variations, or nested inquiries [2]. Subsequent statistical dialogue models and task-oriented architectures incorporated intent classification and slot-filling classifiers; however, their scaling was bottlenecked by extensive manual annotation requirements.

B. Dense Embeddings and Transformer Architectures

The introduction of the Transformer self-attention architecture [3] fundamentally transformed natural language processing. Bidirectional representations derived from BERT [4] demonstrated superior performance over bag-of-words and TF-IDF baselines for conceptual document understanding. Sentence-BERT (SBERT) [6] fine-tuned Siamese networks to map semantically equivalent sentences into contiguous vector spaces, enabling sub-linear nearest-neighbor search via cosine similarity metrics. Dense Passage Retrieval (DPR) [7] further confirmed that learned dense representations significantly outperform sparse inverted-index approaches in open-domain question answering.

C. Retrieval-Augmented Generation (RAG) and Hallucination Control

To mitigate knowledge staleness and catastrophic forgetting in static generative models, Lewis et al. [9] formalized Retrieval-Augmented Generation (RAG). By combining a non-parametric dense retrieval index with a parametric sequence-to-sequence generator, RAG grounds output generation within explicit factual contexts. Comprehensive surveys by Gao et al. [10] and Ji et al. [11] emphasize that conditioning LLM synthesis on retrieved reference passages reduces factual hallucinations by over 70%, rendering generative architectures viable for regulated and institutional applications.

D. Architectural Security and Access Control

Secure enterprise applications require structural authorization boundaries. Sandhu et al. [12] formalized Role-Based Access Control (RBAC), establishing that permissions mapped to organizational roles ensure operational compartmentalization. In modern web architectures, stateless JSON Web Tokens (JWT) [13] offer tamper-evident, cryptographically signed assertion of identity and role claims across distributed RESTful microservices.

TABLE I: SUMMARY OF RELATED RESEARCH DOMAINS

Research Area	Primary Literature	Key Mechanism	Fundamental Limitation
Rule-Based Systems	Weizenbaum [1]	Pattern Matching	Brittle; fails on lexical variation
Conversational Systems	Adamopoulou et al. [2]	Intent-Slot Parsing	Lacks dynamic factual grounding
Pretrained Transformers	Vaswani [3], Devlin [4]	Multi-Head Attention	Parametric memory hallucinations
Vector Representations	Reimers [6], Karpukhin [7]	Siamese Dense Encoding	Retrieval only; no text generation
RAG Frameworks	Lewis [9], Gao [10]	Dense Context Injection	General; lacks role/ticket federation
Security Architectures	Sandhu [12], RFC 7519 [13]	RBAC and JWT Tokens	Generic; requires domain integration

III. RESEARCH GAPS IN EXISTING METHODS

A comparative analysis of the state-of-the-art reveals several architectural gaps in existing administrative helpdesk solutions:

1. Absence of Multi-Tenant Departmental Isolation: Most enterprise helpdesk tools index documents into a single unstructured corpus, frequently causing cross-departmental information leakage (e.g., medical claims rules conflated with travel reimbursement policies).

2. Unverified Generative Output: Commercial AI chatbots lack source-attribution transparency, making it impossible for administrative staff to verify answers against official government gazettes.

3. Disjoint Support Lifecycles: When a chatbot fails to address an inquiry, existing tools terminate the interaction without automated escalation, requiring users to restart their grievance filing process manually.

4. Data Sovereignty Risks: Proprietary cloud APIs transmit internal institutional documents to external servers, violating institutional cybersecurity guidelines.

IV. PROPOSED METHODOLOGY

The system methodology follows an iterative engineering paradigm structured across six operational phases:

TABLE II: METHODOLOGICAL WORKFLOW STAGES

Phase	Title	Operational Scope
Phase 1	Requirements Engineering	Department taxonomy (HR, Finance, IT, Pension, Admin) and

		permission mapping.
Phase 2	Document Ingestion	PDF parsing, normalization, sliding-window chunking (S=800, O=120), dense vector generation.
Phase 3	Semantic Retrieval & RAG	Cosine similarity ranking, prompt assembly, on-premise LLM execution (llama3.2).
Phase 4	Automated Escalation	Anti-hallucination threshold evaluation (theta=0.35, tau=0.60) and ticket creation.
Phase 5	Security Enforcement	Stateless JWT authentication, bcrypt salt encryption, administrative audit logging.
Phase 6	System Validation	Automated pytest suite, Vitest UI testing, end-to-end cloud deployment verification.

V. SYSTEM DESIGN AND IMPLEMENTATION

A. Architectural Overview

The platform utilizes a layered client-server paradigm. The presentation layer comprises a responsive web single-page application (React 18, Vite, Tailwind CSS) and a cross-platform mobile application (React Native, Expo). Client applications interface with an asynchronous backend engine constructed using FastAPI (Python 3.11). Relational data (users, tickets, audit logs, comments) is managed via an enterprise PostgreSQL database, while dense passage vectors are indexed in an embedded ChromaDB instance. Language model inference is handled on-premise through Ollama running Meta Llama 3.2.

B. Core Algorithmic Formulations

1) Text Chunking and Embedding Generation: Uploaded administrative circulars are sanitized and segmented using a sliding window algorithm: Chunk_i = T[i * (S - O) : i * (S - O) + S], where S = 800 characters represents window size, O = 120 characters denotes overlap to preserve cross-boundary semantics, and T is the normalized document text. Each text chunk is mapped into dense 384-dimensional space via SentenceTransformers: v_i = Encoder(Chunk_i) in R^384.

2) Semantic Matching and Similarity Metric: Given an incoming employee query q, the query vector q = Encoder(q) is evaluated against stored chunk vectors v_k belonging strictly to the selected department collection C_dept: Sim(q, v_k) = (q . v_k) / (||q|| ||v_k||). The top-K (K=4) candidate chunks satisfying Sim(q, v_k) >= theta (theta = 0.35) are retrieved.

3) Context Assembly and Grounded Generation: If candidates exist, retrieved chunks are synthesized into an immutable context block C = Union(Chunk_k). The prompt supplied to the local inference engine is structured as: Prompt = SystemPrompt || C || q. If max Sim(q, v_k) < 0.60, the system generates a tentative answer while concurrently asserting suggest_ticket = TRUE. If similarity fails to achieve the floor threshold theta = 0.35, generative inference is suppressed, and the system prompts immediate support ticket generation.

TABLE III: MATHEMATICAL NOTATIONS AND OPERATIONAL PARAMETERS

Parameter	Symbol	System Value	Technical Description
Chunk Size	S	800 chars	Fixed character window length
Overlap Length	O	120 chars	Sliding overlap to preserve context
Embedding Dim	D	384	Output dimension of all-MiniLM-L6-v2
Top-K Candidates	K	4 chunks	Maximum retrieved context segments
Minimum Cutoff	theta	0.35	Cosine threshold for candidate retrieval
Confidence Floor	tau	0.60	Threshold below which tickets are suggested

C. Human-in-the-Loop Ticket Resolution Pipeline

When an automated query is deemed inconclusive or requires physical administrative authorization (e.g., equipment procurement, leave sanction), the request transitions into the ticketing engine. The system generates an immutable identifier conforming to TKT-YYYY-XXXXXXXX (where Y is the current calendar year and X is a pseudorandom alphanumeric token). The lifecycle follows a strict finite state machine: S in {Open, InProgress, Resolved, Closed}. Administrative permissions allow any designated Department Officer, Department Admin, or Super Administrator to review submitted tickets, post official resolution guidance, append chronological status updates, and archive completed cases.

VI. RESULTS AND DISCUSSION

A. Experimental Environment

Experimental evaluation was conducted across two discrete execution profiles: (1) Local On-Premise Environment: Windows 11 OS, AMD Ryzen / Intel Core i7 architecture, 16 GB DDR4 RAM, Ollama runtime running quantized llama3.2 (3-billion parameter model), and local SQLite/PostgreSQL; and (2) Production Cloud

Infrastructure: Global edge deployment utilizing Vercel CDN for the React SPA, Render Linux Container (Python 3.11 runtime) for the REST API gateway, and AWS Singapore (ap-southeast-1) managed PostgreSQL via Supabase connection pooling.

TABLE IV: QUANTITATIVE SYSTEM PERFORMANCE AND LATENCY METRICS

Performance Parameter	Target SLA	Measured Local	Measured Cloud	Evaluation Criterion
Chat Response Latency (LLM Active)	< 10.0 s	5.0 - 8.0 s	Model Offloaded	Full RAG pipeline inference
Fallback Search Latency (Lexical Mode)	< 2.0 s	0.4 - 0.8 s	0.4 - 1.2 s	Lightweight vector-to-text fallback
Document Indexing Latency (500 words)	< 30.0 s	14.2 s	15.0 s	Ingestion, chunking, and ChromaDB insert
Ticket Listing Query (100 records)	< 2.0 s	0.8 s	1.0 s	REST API query execution and UI render
Dashboard Analytics Aggregation	< 3.0 s	1.2 s	1.5 s	Complex SQL join and metric aggregation
Concurrent Stress Capacity	100 users	100 users	100 users	Simulated parallel clients without failure

B. Retrieval Precision and RAG Effectiveness

The system was pre-seeded with 15 verified regulatory circulars across 5 departments (HR, Finance, IT, Pension, Administration). The retrieval engine was evaluated against 50 representative benchmark queries covering leave rules, travel allowances, digital life certificates, and workstation allocations.

TABLE V: RAG KNOWLEDGE RETRIEVAL & GROUNDING EVALUATION

Metric Parameter	Measured Value	Evaluation Protocol
Configured Departments	5	Administrative operational units
Indexed Policy Documents	15 circulars	3 comprehensive official guidelines/department
Mean Cosine Confidence Score	0.78	Evaluated on valid domain inquiries
Positive User Feedback Rate	85.0%	Binary telemetry via thumbs-up/down feedback
Test-Set Hallucination Rate	0.0%	Zero ungrounded responses when threshold applied
Fallback Engagement Rate	100.0%	Unindexed queries correctly escalated to tickets

TABLE VI: SAMPLE TEST QUERIES, GROUNDING SOURCES, AND VERIFICATION

Department	Submitted Test Query	Grounding Document Cited	Cosine Score	Verification Result
HR	What is the annual earned leave entitlement?	HR_Leave_Rules_2024.pdf	0.84	Factual leave days correctly cited
Finance	How do I submit Form 12BB for tax savings?	Finance_Tax_Declaration_Guide.pdf	0.81	Chronological workflow outlined
IT	How do I reset my official NIC email password?	IT_Email_Security_SOP.pdf	0.88	Self-service URL and policy quoted
Pension	What is the eligibility for Family Pension?	Pension_Scheme_Guidelines.pdf	0.79	Dependent thresholds verified
Admin	What are the standard office working hours?	Admin_Office_Regulations.pdf	0.92	Core timings and lunch break cited
Cross/Out	Book flight tickets for personal leisure travel.	None (Suppressed)	< 0.35	Declined; ticket creation suggested

C. System Verification and Acceptance Testing

Automated testing protocols confirmed system integrity: (1) Backend Suite: 19 automated pytest cases executed with 100% pass rate (test_auth.py, test_chat.py, test_rag.py, test_tickets.py); (2) Frontend Suite: 5 Vitest component test suites validating auth context guards, token storage, and error boundaries; (3) End-to-End Suite: The automated smoke_test.py pipeline executed sequential operational verification covering /health, /auth/login, /departments/,

/chat/ask, and /tickets/ POST endpoints with zero operational failures; and (4) Comprehensive Audit: 33/33 discrete acceptance test criteria were satisfied across authentication, ticket lifecycle, administrative oversight, and mobile client access.

VII. LIMITATIONS

While the architecture provides robust enterprise automation, several operational limitations are acknowledged: (1) Linguistic Coverage: The prototype currently operates on monolingual English corpora. In multi-lingual administrative settings across India, integration of Hindi and regional languages is necessary; (2) Local Compute Constraints: High-throughput on-premise execution of quantized LLMs requires localized workstation GPUs. On CPU-only nodes, inference latencies can extend during peak loads; and (3) Lack of Real-Time ERP Binding: The system functions over static document indexes and does not execute transactional mutations within legacy government HRMS or integrated financial management systems.

VIII. CONCLUSION AND FUTURE WORK

This paper presented an AI-based multi-department helpdesk portal engineered for public sector governance. By synthesizing dense vector retrieval with on-premise generative models and an automated ticketing lifecycle, the system delivers verifiable, document-grounded responses while establishing a human escalation bridge for administrative resolution. The implementation of strict four-tier RBAC, cryptographic token verification, and automated audit logging guarantees compliance with data privacy mandates.

Future research will focus on extending semantic indexing to support Hindi and 22 scheduled Indian languages via multilingual embedding models (e.g., IndicBERT), implementing voice-driven natural language interfaces for accessibility, incorporating Optical Character Recognition (OCR) pipelines to parse scanned archival circulars, and developing federated API connectors for secure integration with state e-governance platforms such as DigiLocker, UMANG, and HRMS.

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